A BRIEFING FOR ESRI

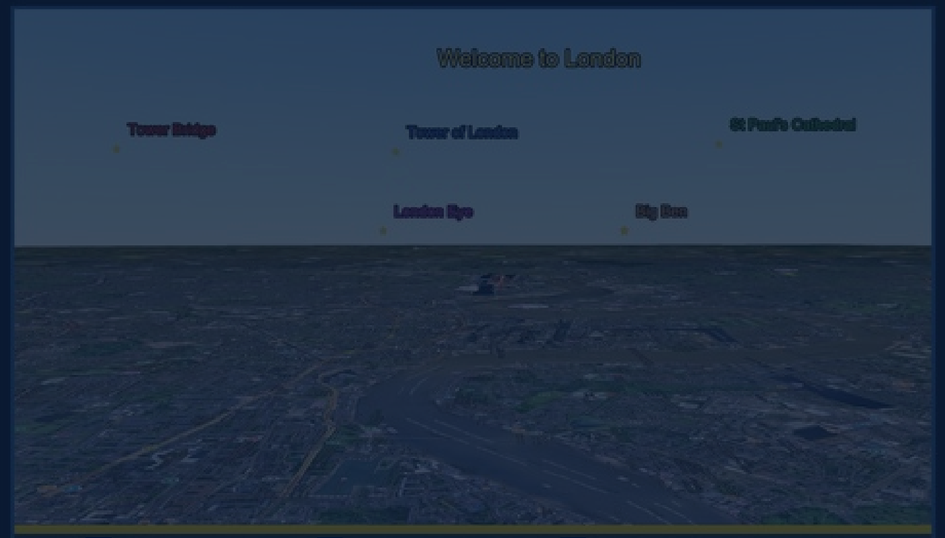
ArcGIS Earth Unlocked

Seven free projects that add five missing capabilities to ArcGIS Earth, built and field-tested by a fire-service radio engineer.

Jim G · github.com/Jim-dc95811 · November 2026

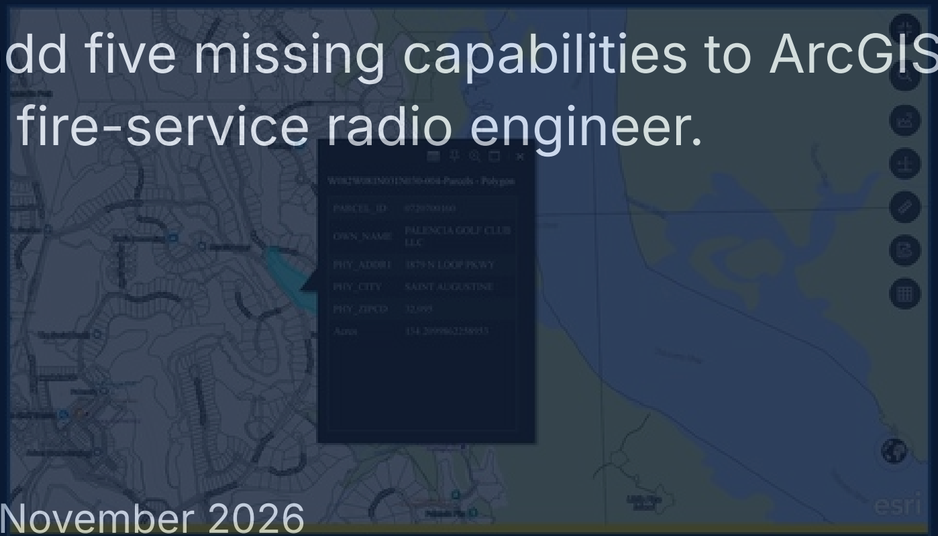


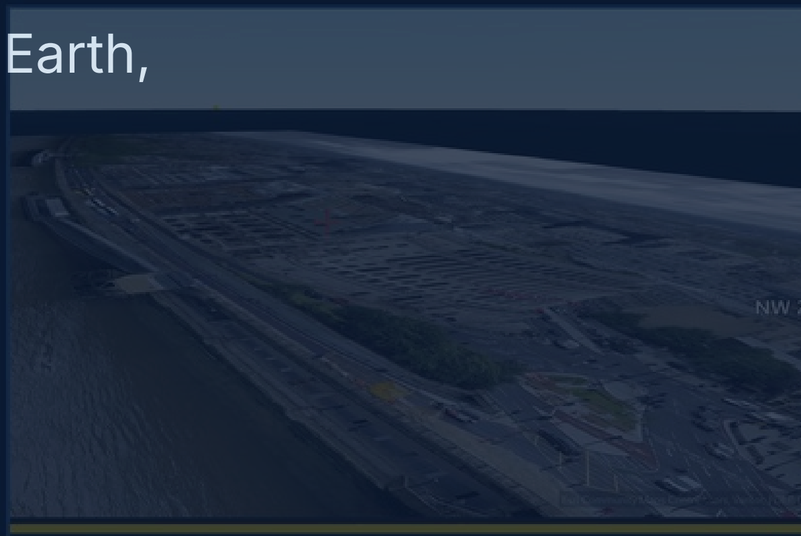












POINTLAYER
hydrants and public points, offline

6 PARCEL100
click a parcel: owner and address

7 PICTURE WORLD MAKER
fly into giant photographs

THE GAP

Five things ArcGIS Earth can’t do on its own

Capability	ArcGIS Earth today	Added by
Offline map-making	Opens offline TPKX; making one usually takes ArcGIS Pro	mb2tpkx, Master 4, Batch Factory
Joystick flight	Mouse, keyboard, touch, SpaceMouse	EarthStick
City-to-city travel	Search a place, the camera flies there	World Gates, World Gate Maker
Easy object authoring	Draw tools and KML import	Object Factory
Picture worlds	A plain photo has no place on the globe	Picture World Maker

WHERE IT CAME FROM

First responders needed sharp satellite maps with no signal. The tracking system they used stopped at zoom 17.

Built in the field

37 years of electronics and radio work at a state fire department, applied to mapping.

Built with AI

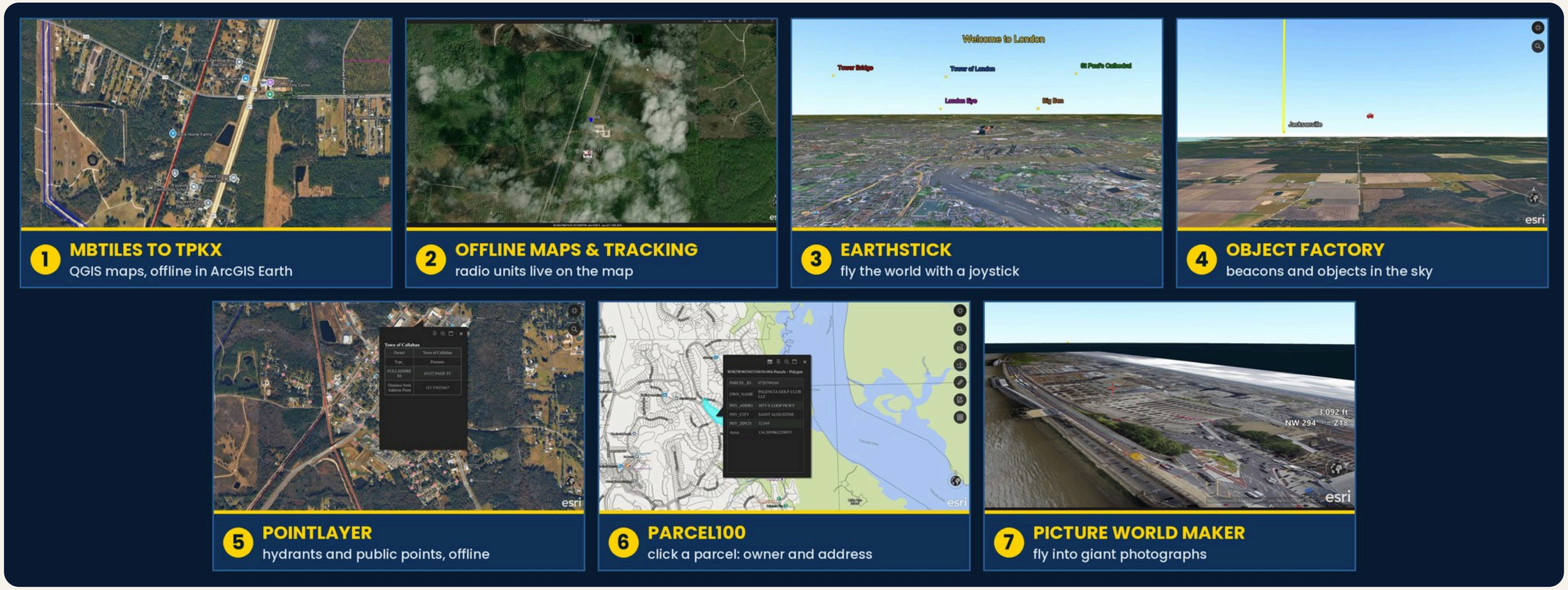
Conceived, directed and tested by a non-programmer, written with ChatGPT and Claude.

Given away

Free and public on GitHub, written for people with no GIS training.

THE FAMILY

Seven projects, one renderer



1 · OFFLINE MAPS

Make Z20 offline maps without ArcGIS Pro

40 GB-class

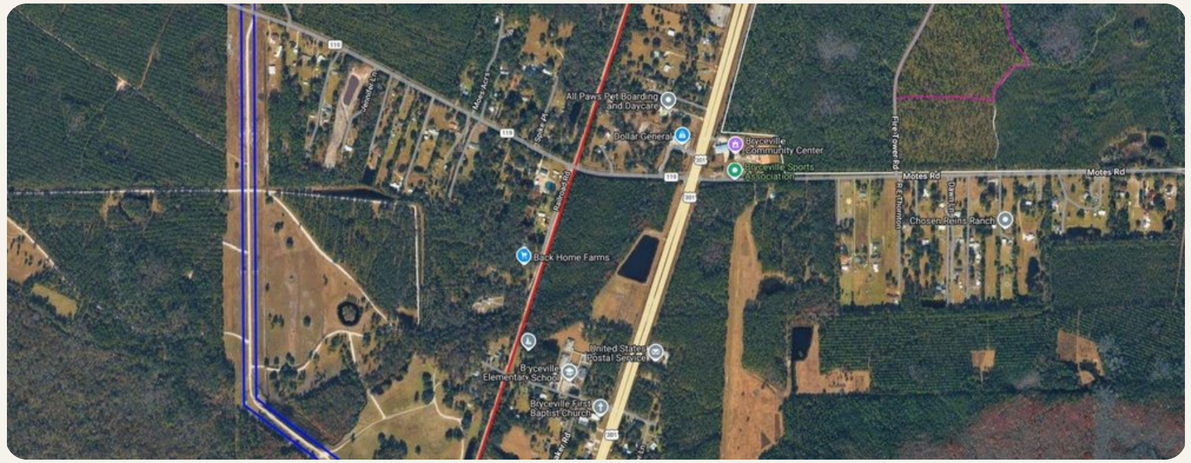
TPKX packages flown in ArcGIS Earth

100 cells

Master 4 splits a 1° box into numbered 0.1° tiles

QGIS → TPKX

mb2tpkx converts MBTiles; Batch Factory serves Pro users



2 · JOYSTICK FLIGHT

Fly ArcGIS Earth with a flight stick

Automation API

EarthStick drives the camera on localhost:8000

Pilot readout

Height, heading and equivalent zoom, even full screen

No extra libraries

Plain Python, one file to click



3 · CITY TO CITY

Fly through a gate, arrive in another city

22 World Gates

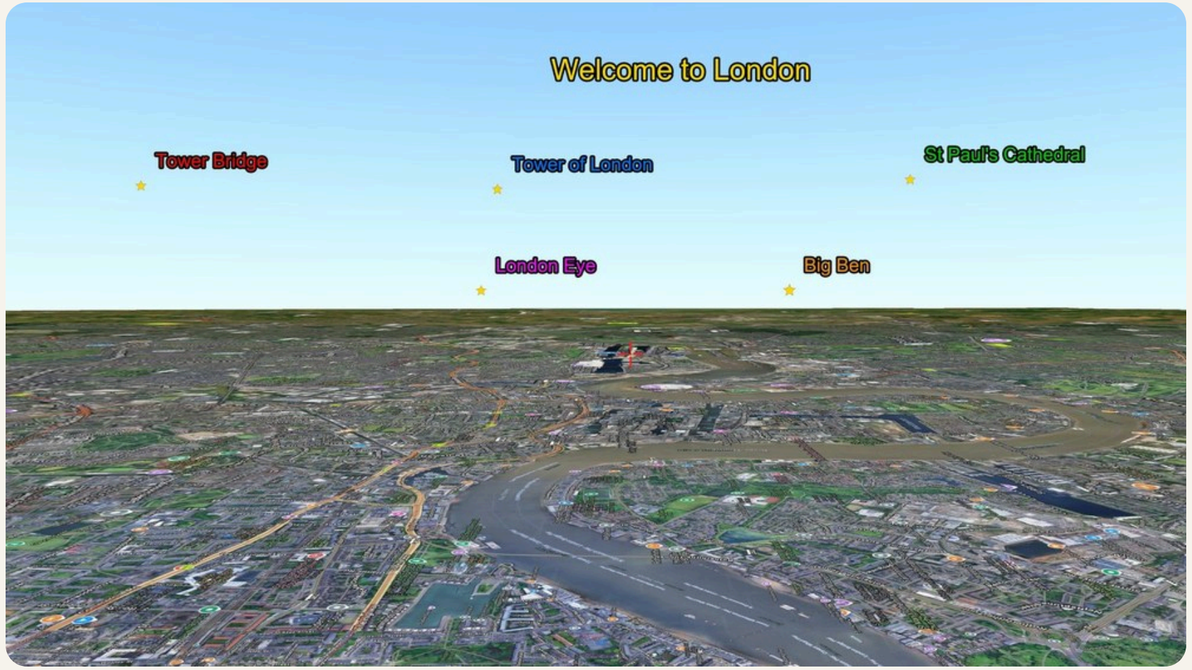
Jacksonville to an Atlanta hub and seven world cities

Lined up for you

A height light and tractor beam centre the pilot

Your own gates

World Gate Maker links any two places



4 · OBJECTS

Place clickable objects in the sky in seconds

11 object types

Beacons, staging, water, medical, a sky compass

Live camera popup

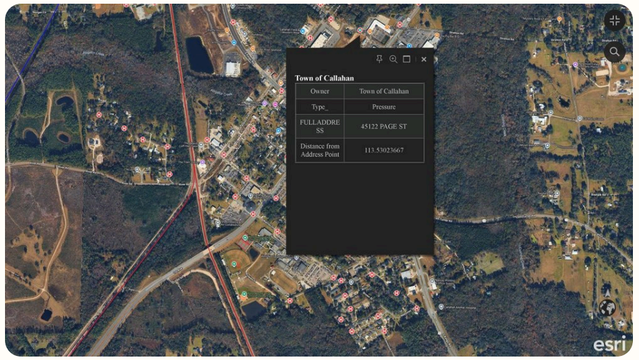
A traffic camera that refreshes inside ArcGIS Earth

Hide above

Objects vanish past a set height, calibrated on screen

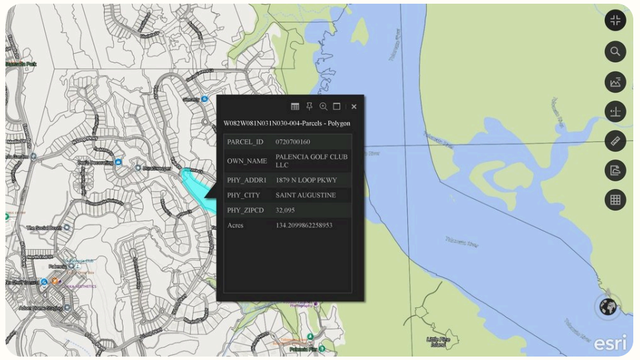


Public data and live units, offline



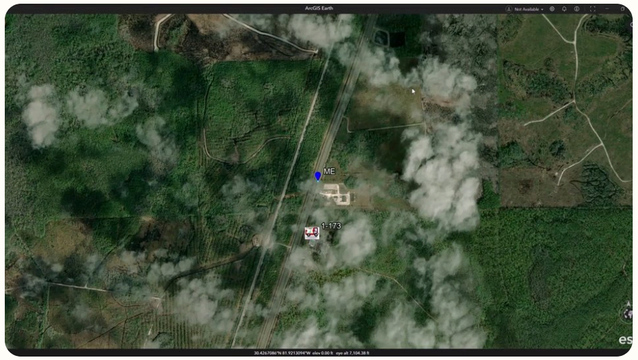
PointLayer

30,688 published hydrants in one county, saved as one offline KMZ



Parcel100

Click any parcel for owner and address

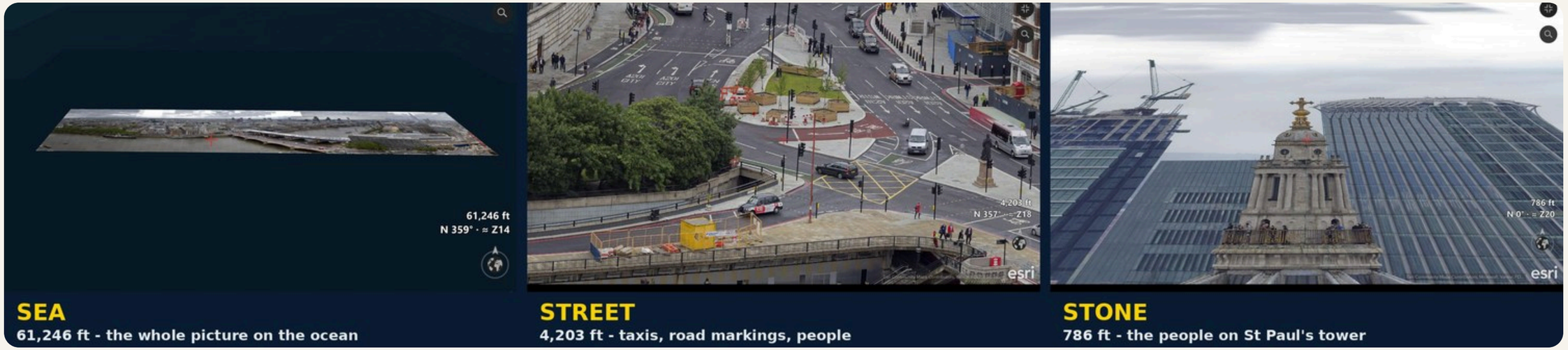


PRAVE/ME Tracker

Radio GPS units drawn live on the map

6 · PICTURE WORLDS

Any giant picture becomes a place you can fly into



SEA
61,246 ft - the whole picture on the ocean

STREET
4,203 ft - taxis, road markings, people

STONE
786 ft - the people on St Paul's tower

50,000 px
panorama, about 600 megapixels

55 seconds
to build 35,216 tiles

No GIS needed
pick a picture, get a TPKX

FIELD NOTES · ARCGIS EARTH DESKTOP 3.0.1

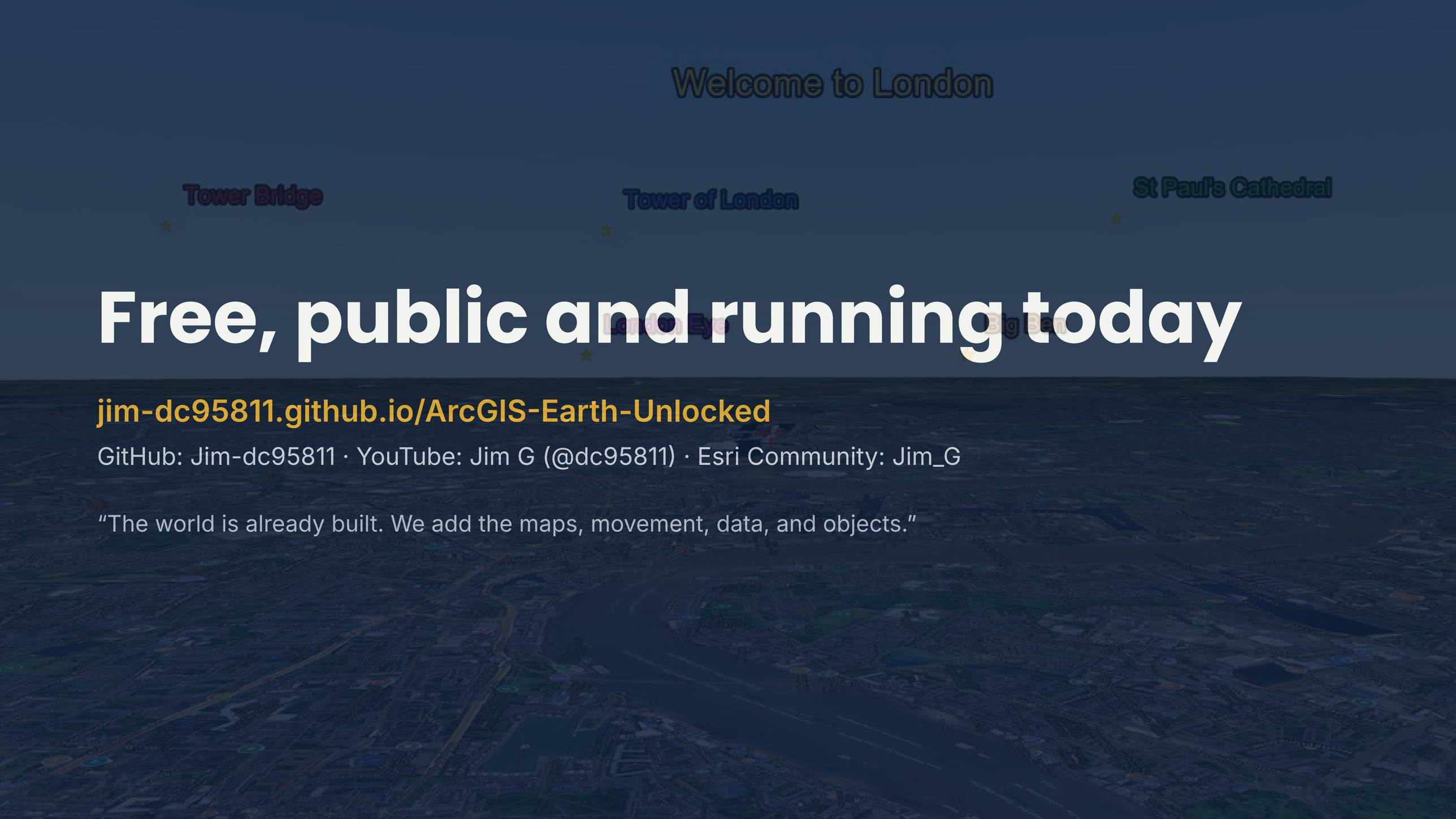
What building on ArcGIS Earth taught us

Measured	What it means
Popups run JavaScript	Objects can show live cameras and call the local API
The camera API is fast	Fast enough to fly smoothly with a joystick
KML Camera and LookAt are ignored	A link always lands straight overhead
API graphics ignore height	Shapes drape on the ground, never in the sky
Re-added drawings flash	No smooth moving lines or live tracks in the sky

SUGGESTIONS

Five things ArcGIS Earth could add natively

1. Flight stick and gamepad input, beside the SpaceMouse
2. Honour KML Camera and LookAt views
3. Move an API drawing without the highlight flash
4. Show eye altitude in full-screen mode
5. Open MBTiles directly, or convert them to TPKX



Free, public and running today

jim-dc95811.github.io/ArcGIS-Earth-Unlocked

GitHub: Jim-dc95811 · YouTube: Jim G (@dc95811) · Esri Community: Jim_G

“The world is already built. We add the maps, movement, data, and objects.”